HW1: VAE for Chest X-ray Generation

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Abstract

This assignment utilizes a Variational Autoencoder (VAE) model for the task of chest X-ray image generation. The dataset consists of over 5,000 chest X-ray images (including both normal and diseased cases) sourced from Kaggle. The goal after training is to produce clear and recognizable X-ray images. Fréchet Inception Distance (FID) and Inception Score (IS) are used as the evaluation metrics for this task. https://github.com/evanchiang720/Deep-Generative-Model/tree/main/hw1

1 Introduction and Method Formulation

Variational Autoencoders (VAEs) are a well-established generative technique. They employ a neural network architecture with an Encoder and a Decoder, enabling the model to learn how to map images into a latent space and reconstruct them back into images. VAEs can generate a wide variety of images depending on the training data; for this project, medical imaging was chosen.

1.1 Objective Function (ELBO)

The Evidence Lower Bound (ELBO) consists of two components: Reconstruction Loss and KL Divergence. The Reconstruction Loss evaluates whether the generated images closely resemble the training images, while the KL Divergence encourages diversity in the generated outputs by forcing the latent distribution to closely approximate a standard normal distribution.

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{MSE}}(x, \hat{x}) + \beta \cdot D_{\text{KL}}\left(q_\phi(z|x) \,\|\, p(z)\right) \tag{1}$$

In the implementation, the reparameterization trick is used to overcome the non-differentiability of the random sampling step. By decoupling the stochasticity from the learnable parameters, sampling is expressed as $\mu + \epsilon \times \sigma$, where ϵ is sampled randomly from a standard normal distribution, and μ and σ are outputs produced by the model.

1.2 Model Architecture

The Encoder and Decoder architectures utilize convolutional and transposed convolutional (deconvolutional) layers, respectively. Each side consists of 3 layers, with channels increasing from 1 to 128 before reaching a latent space dimension of 256, and then gradually reducing back to 1. The kernel size is set to 4 across the layers. ReLU is used as the primary activation function, while a Sigmoid activation function is applied at the final layer to bound the pixel values between 0 and 1.

2 Experimental Setup and Hyperparameters

The model was trained in a Conda environment using an NVIDIA RTX 3050 Ti GPU. To avoid Out-Of-Memory (OOM) errors, the image resolution was restricted to 64×64. During data preprocessing, all images were resized to 64×64 and converted to grayscale. Below are the hyperparameter settings used for training:

39th Conference on Neural Information Processing Systems (NeurIPS 2025).

Table 1: Hyperparameter settings used for VAE training.

Hyperparameter	Value	Hyperparameter	Value
Input Resolution	64x64x1	Optimizer	Adam
Latent Dimension (z)	256	Learning Rate	1e-3
Batch Size	64	Reconstruction Loss	MSE
Total Epochs	50	KL Weight (β)	0.001

3 Results and Evaluation

3.1 Qualitative Results: Loss and Samples

The training loss converged rapidly during early epochs and stabilized in later epochs (Figure 1). For generated images, although not extremely sharp, structural contours of the thoracic cavity and ribs are clearly recognizable. Furthermore, structural diversity is observable across samples—some depict a largely bright thoracic region, while others appear almost entirely dark (Figure 2).

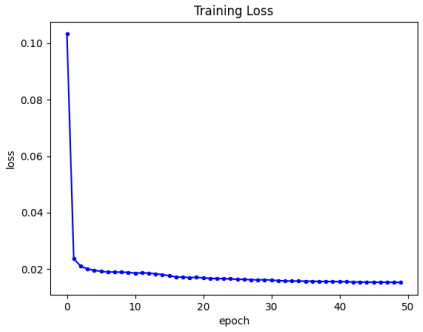


Figure 1: Training loss curve across epochs.

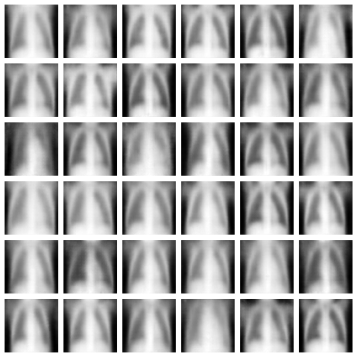


Figure 2: Grid of generated chest X-ray samples.

3.2 Quantitative Metrics (FID and IS)

PyTorch-Ignite metrics were used for evaluation. The FID measures the feature distribution distance relative to real images (lower is better), while the IS measures the clarity and diversity of generated images (higher is better). Both FID and IS scores achieved expected standard levels (Table 2).

Table 2: Quantitative generative evaluation results.

Metric	Score	Target / Criteria
Fréchet Inception Distance (FID)	0.1808	Lower is better
Inception Score (IS)	1.2874	Higher is better

3.3 Discussion and Analysis

The generated images appear somewhat blurry due to the inherent characteristics of VAEs. Specifically, the Mean Squared Error (MSE) loss minimizes pixel-level variance by averaging predictions, which inherently smoothes out fine details. Additionally, the IS score is lower compared to natural image benchmarks on ImageNet (which exhibit much higher semantic diversity across 1,000 distinct classes). This occurs because chest X-rays are structurally homogeneous and limited to grayscale channels. Potential solutions to these limitations include adopting a Generative Adversarial Network (GAN) architecture—where a discriminator enforces sharpness—and applying data augmentations like rotation or flipping during training to enhance generative diversity.

4 Conclusion

In this report, I implemented and trained a convolutional VAE for chest X-ray generation, achieving results that met initial expectations. The project highlighted both the strength of VAEs—generating realistic images with only a few thousand training samples—and their main drawback, namely the tendency to produce blurry images.